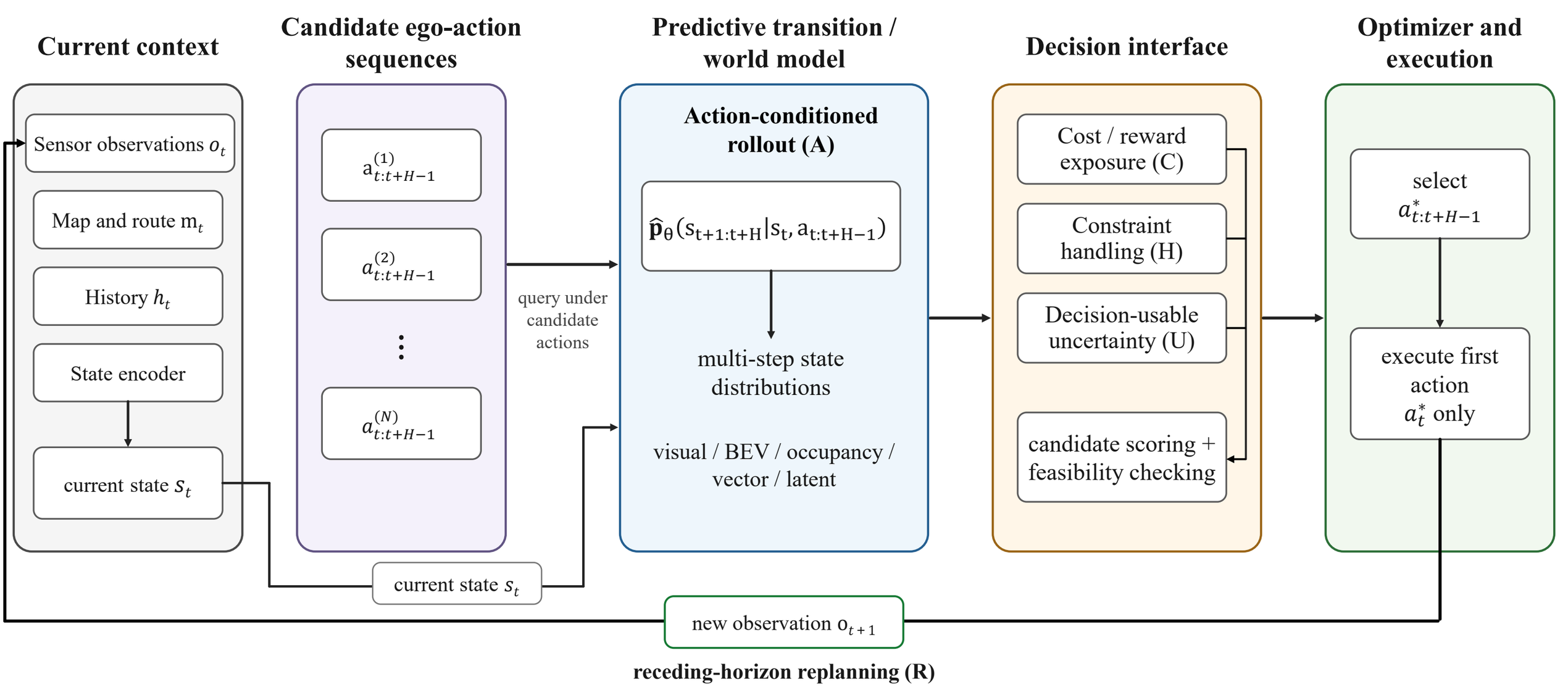

Current context
Sensor observations o_t
Map and route m_t
History h_t
State encoder
current state s_t
Candidate ego-action sequences
$a^{(1)}_{t:t+H-1}$
$a^{(2)}_{t:t+H-1}$
$a^{(N)}_{t:t+H-1}$
query under candidate actions
Predictive transition / world model
Action-conditioned rollout (A)
$\hat{p}_\theta(s_{t+1:t+H}|s_t, a_{t:t+H-1})$
multi-step state distributions
visual / BEV / occupancy / vector / latent
Decision interface
Cost / reward exposure (C)
Constraint handling (H)
Decision-usable uncertainty (U)
candidate scoring + feasibility checking
Optimizer and execution
select $a^*_{t:t+H-1}$
execute first action a^*_t only
current state s_t
new observation o_{t+1}
receding-horizon replanning (R)